

Nirma University
Institute of Technology
Semester End Examination (IR), January-2023
M.C.A, Semester- I
6CS101 Object Oriented Programming

Roll/
Exam No.

Supervisor's initial
with date

Time: 3 Hours

Max. Marks: 100

Instructions:

1. Attempt all questions.
2. Figures to right indicate full marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.
5. Assume suitable data wherever necessary and mention the same.

SECTION-I

- Q-1** **Answers the following:** [18]
(A) Distinguish between Java Development Kit (JDK), Java Runtime Environment (JRE) and Java Virtual Machine (JVM)? [6]
CO1,BL4
(B) Define tokens in java. Explain the various types of java tokens [6]
CO1,BL2 in detail.
(C) Why 'Switch Statement' is used in java programming? Also [6]
CO1,BL3 write a suitable example which show the use of the 'Switch Statement'.
- Q-2** **Answers the following:** [16]
(A) Explain various Relational operators. Also give an example of [8]
CO2,BL4 relational operators using java programming.
(B) Explain the String class. Also show the use of the equals(), [8]
CO2,BL3 concat() and substring() methods with a suitable example.
- Q-3** **Answers the following:** [16]
(A) Identify the use of nesting of loop. Also write a program to print [8]
CO2,BL6 the following output using nesting of loop.
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
(B) Briefly describe Constructor. Differentiate between the Java [8]
CO2,BL4 Constructor and Java method by giving a suitable example.

SECTION-II

- Q-4** **Answers the following:** [18]
- (A) What is hierarchical inheritance? Also write an example of hierarchical inheritance. (6)
CO2,BL3
- (B) What do you mean by JDBC (Java Database Connectivity) and JDBC Drivers? How JDBC is useful in order to develop the Client Server application. (6)
CO3,BL4
- (C) Differentiate between Method overloading and Method overriding by giving suitable example. (6)
CO2,BL4
- Q-5** **Answers the following:** [16]
- (A) What is the relationship between the static variable and static method? Also write an example which will show the use of them. (8)
CO2,BL5
- (B) Discuss the function of interface with a suitable example. (8)
CO2,BL6
- Q-6** **Answers the following:** [16]
- (A) Is it compulsory to use exception handling in Java programming? Explain the try and catch block with a suitable example which will use an exception. (8)
CO3,BL4
- (B) What is a stream in context of Java I/O? What types of streams does Java define? Write a java program which will read the data from the file and print on the screen. (8)
CO3,BL6